Predicting three-dimensional pharmacophore fingerprints from constitution alone

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Technical report · PharmCast

ABSTRACT

Pharmacophore fingerprints describe which arrangements of chemical features a molecule can present in three dimensions, and they are among the most useful descriptors available for scaffold hopping, focused library design and similarity searching. Their cost is dominated not by the fingerprint but by the conformer ensemble it requires: almost all of the cost of fingerprinting one catalog molecule over 100 conformers is conformer generation, and only a small fraction of it is the fingerprint calculation itself. We describe *PharmCast*, a feedforward model that predicts the complete 10,549-bit ensemble fingerprint directly from the molecule's two-dimensional structure, with no conformer generation at any stage. Trained on 5,887,229 catalog molecules whose ensemble fingerprints were computed conventionally, PharmCast reproduces the held-out fingerprint at a median per-molecule Matthews correlation of 0.881. On the quantity that is actually consumed downstream, the pharmacophoric similarity between two molecules, it agrees with the reference calculation at Pearson 0.980 with a median absolute error of 0.008, and it orders candidate pairs correctly 94.1% of the time overall and 99.71% when the true difference exceeds 0.10. A complete comparison of two molecules runs from structure alone, with no conformer generated at any point. We report where the approximation holds, where it degrades, and a measurement of the reference calculation's own reproducibility that bounds what any surrogate could achieve.

Keywords: pharmacophore fingerprint, PharmPrint, PolyPharmPrint, conformer ensemble, virtual screening, scaffold hopping, surrogate model.

1. Introduction

A pharmacophore fingerprint encodes the three-dimensional arrangements of chemical features that a molecule is capable of presenting. In the PharmPrint formulation [1,2] the basis is a set of three-point pharmacophores: every legal triangle of three typed features held at three binned distances contributes one bit. Features are assigned from constitution rather than from a docked pose, and comprise acceptor, donor, negative, positive, hydrophobic, aromatic and other. Distances are binned into six ranges spanning 2.0 to 24.0 Å. The resulting vector has 10,549 bits.

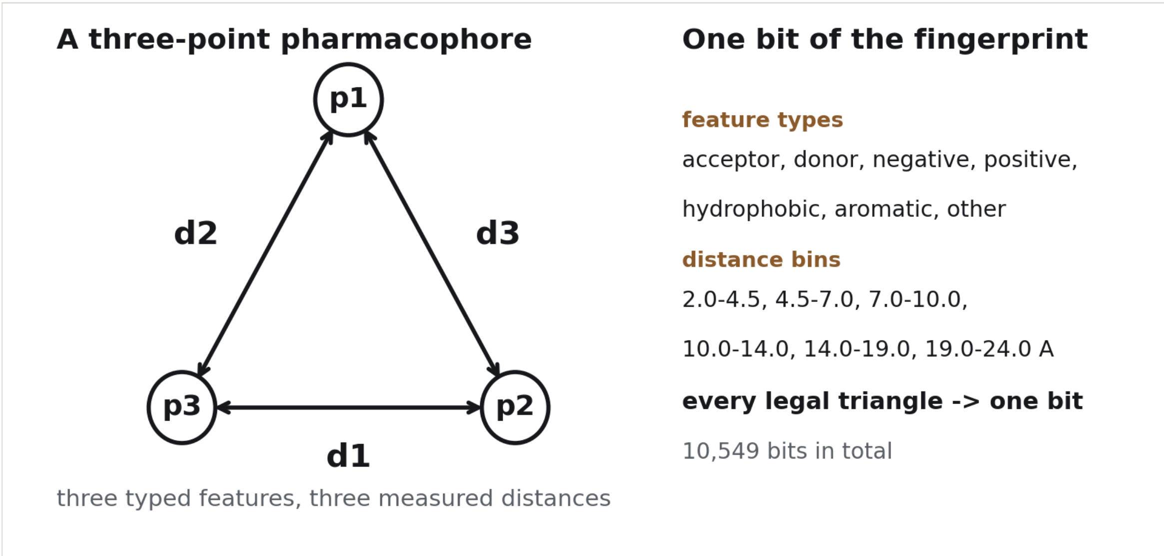

Figure 1. The unit of the descriptor. Three typed features held at three measured distances form one three-point pharmacophore; binning the distances makes it a discrete object, and each legal combination of three types and three bins is one bit of the fingerprint.

Because a single conformer presents only one arrangement, the descriptor is made conformationally honest by building an ensemble. Two conventions are in use. The original PharmPrint approach builds many conformers, fingerprints each, and ORs the bits into one combined vector, which answers what a molecule *can* present across its accessible landscape. PolyPharmPrint instead keeps every conformer's fingerprint separate, so that a match is between a specific conformer of the query and a specific conformer of the hit, preserving the matching geometry. This work targets the first convention: one ensemble fingerprint per molecule, ORed over 100 conformers.

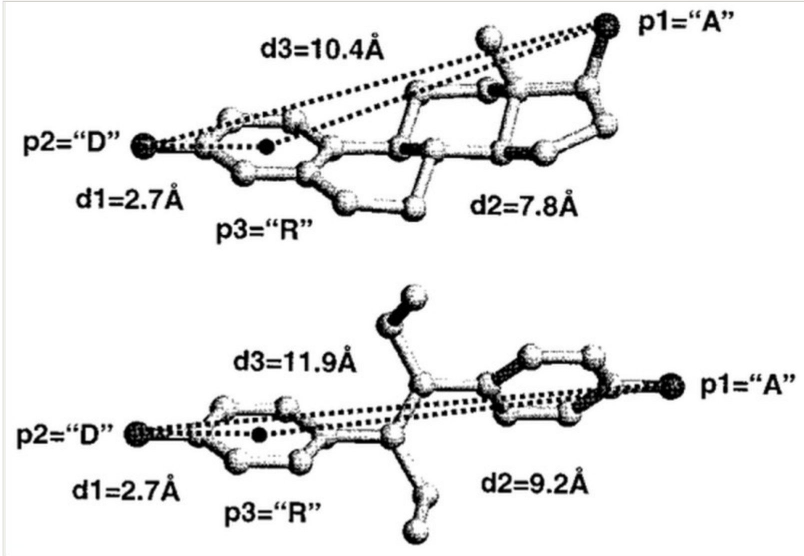

Figure 2. A mapping of one three-point pharmacophore onto (i) estradiol (top), the natural ligand, and (ii) diethylstilbestrol (bottom). The hydroxyl groups are both hydrogen bond donors and acceptors (D, A), and the centroid of the aromatic ring is shown (R). The two molecules share no scaffold, yet present the same three features at distances that fall in the same bins, so the same bit is set in both. Reproduced from reference 1.

The ensemble is where the cost lives. Over 100-conformer builds, conformer generation and optimization dominate the total, while the fingerprint calculation itself is a small fraction of it. The descriptor is therefore far cheaper than the geometry it is computed on. That imbalance is what motivates a surrogate: if the ensemble fingerprint can be predicted from constitution, the entire conformational stage can be skipped for any application that consumes similarities rather than geometries.

We report such a model. It is not a replacement for the conventional calculation, which remains the arbiter for anything acted upon; it is a filter that allows far more chemistry to be placed in front of that arbiter.

2. Methods

2.1 Reference calculation

Ground truth throughout is the conventional route. For each molecule, 100 conformers are generated by distance geometry using experimental torsion-angle preferences (the ETKDG method [3], version 3 [4]), starting from the isomeric line notation for the structure, with hydrogens added and the largest fragment retained. Each conformer is then relaxed against the Universal Force Field [5], as implemented in RDKit [7], and the resulting ensemble is passed through the reference fingerprint implementation. The per-conformer fingerprints are

ORed into one 10,549-bit ensemble vector. The same recipe, with the same fixed seed, is used for every training molecule and every evaluation molecule, so the surrogate is never asked to reproduce a protocol it was not trained on.

2.2 Training data

Training molecules are drawn from a commercial screening collection [8] of 4,612,044 compounds. The collection is broad in scaffold terms, with 135,768 distinct Bemis-Murcko scaffolds in a 200,000 molecule sample of which 87.6% appear exactly once, and simultaneously dense in close analogs, which is characteristic of enumerated catalog chemistry. The model reported here was trained on 5,887,229 molecules.

2.3 Peptide corpus, and the distinction between an observed and a proliferated presentation

The second corpus is drawn from experimentally determined protein structures [10], and it exists to extend the model beyond catalog chemistry into a class of molecule that is unquestionably real. The Protein Data Bank [10] was queried at a resolution limit of 2.5 Å and a free R-factor limit of 0.22, which returned **71,018** entries. Of these **70,818** parsed and **69,450** yielded at least one usable loop.

A loop is defined as a run of two to six residues lying between two annotated elements of secondary structure. Extraction found **1,524,535** loop instances across **135,408** chains, a median of fifteen per entry. Each is capped using the real flanking atoms present in the structure rather than with invented groups, so the capped molecule remains a description of something observed. Backbone continuity is enforced by requiring the carbon to nitrogen distance across each junction to fall between 1.0 and 2.0 Å; this check rejected **54,520** candidates that would otherwise have carried chain breaks, alternate conformations or missing density into the corpus, and it is not optional. Collapsing the survivors gives **117,741** distinct sequences and **133,136** distinct capped molecules.

The same sequence frequently appears many times across the Protein Data Bank, and its instances are not equivalent. This is worth stating plainly because the opposite is widely assumed: for short loops, sequence does not fix structure. A loop of a given sequence adopts different backbone geometries in different structures, in different chains of the same structure, and in different crystal forms, and each of those geometries presents a different arrangement of pharmacophoric features. Measured across **1,481** sequences observed three or more times, the median pharmacophoric agreement between two observations of the *same* sequence is **0.878**, with 18% of comparisons below 0.70 and 19% effectively identical. Sequence identity therefore does not imply pharmacophoric identity, and this spread is signal rather than noise.

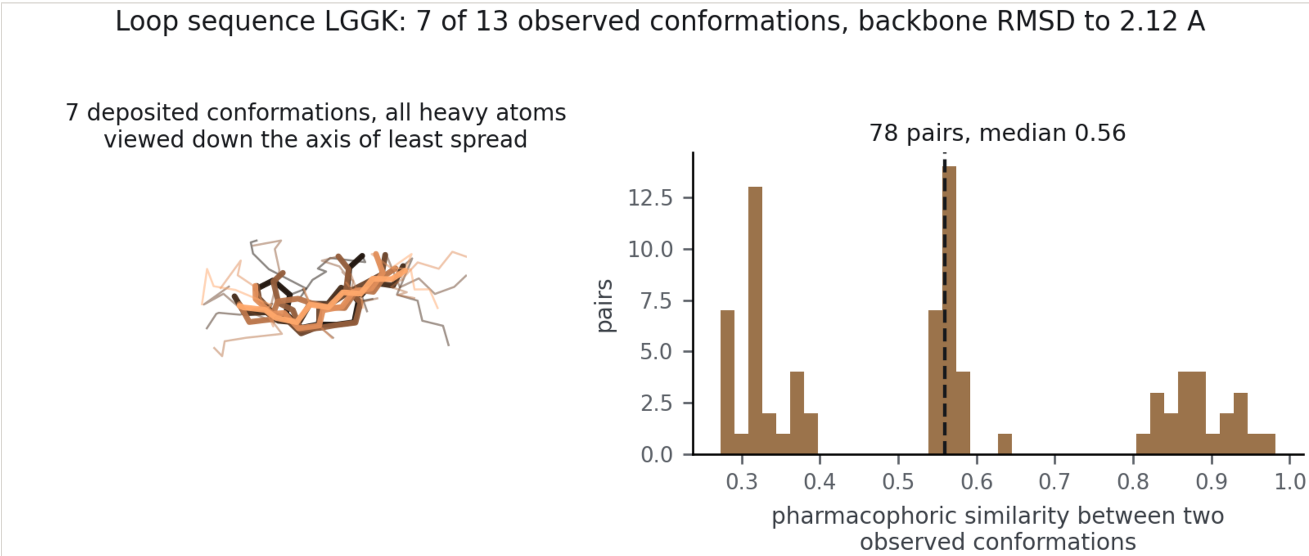

Figure 3. Sequence does not determine pharmacophoric presentation. Searching 19,108 loop sequences for the largest divergence between their own observed conformations returns LGGK, shown here as observed in **13** independent Protein Data Bank entries, all of them determined at 1.8 Å or better with a free R-factor at or below 0.20. Left, 7 of those conformations drawn with all heavy atoms and superimposed on the backbone; they disagree by 2.12 Å of backbone root-mean-square deviation. Right, the pharmacophoric similarity between every pair of the observed conformations, 78 pairs, with a median of 0.56, of which 72% fall below 0.70. The distribution separates into distinct populations with gaps between them, so this sequence does not adopt one presentation with noise around it, it adopts several. Across the search, backbone deviations reach 3.1 Å and all-atom deviations 6.3 Å. The conformations drawn are the most pharmacophorically distinct rather than the best resolved, since selecting by resolution returns the same crystal form repeatedly.

The obvious objection is that this variation is an artifact of crystallographic quality: poorly resolved or loosely refined loops would fit badly and appear to differ. That objection is testable, and it fails. Repeating the search over only those instances determined at 1.8 Å or better with a free R-factor of 0.20 or below leaves 9,218 sequences, and the largest backbone divergence found is **3.11 Å**, unchanged from the unrestricted search. All-atom divergence reaches 5.23 Å. The sequence shown in Figure 3 is drawn entirely from that restricted set: its 13 conformations have a median resolution of 1.65 Å, the best at 1.00 Å, and a median free R-factor of 0.189. These are well-determined structures that genuinely disagree.

Table 1. The most conformationally divergent loop sequences, restricted to instances at 1.8 Å or better with a free R-factor of 0.20 or below.

SEQUENCE	CONFORMATIONS	BACKBONE RMSD	ALL-ATOM RMSD	MEDIAN RESOLUTION	MEDIAN R-FREE
SGGGG	35	3.11	3.80	1.55	0.163
LGGK	49	2.59	4.38	1.65	0.189
GSGL	34	2.50	4.05	1.54	0.185
GGGK	12	2.43	5.23	1.50	0.159
GPSG	32	2.38	2.75	1.47	0.182
AGGK	9	2.36	4.32	1.77	0.196
DLAV	12	2.34	3.97	1.50	0.188
AGAV	13	2.27	3.07	1.50	0.180

Two fingerprints can consequently be defined for a peptide, and the difference matters.

The **rigid** fingerprint is computed on a single conformation exactly as it was observed, with no conformer generation of any kind. Every pharmacophore it records was physically present in a crystal structure. It answers what that loop *was* doing in that context, and it is the more conservative object: it cannot contain an arrangement that was never seen.

The **proliferated** fingerprint is the union of two things. It ORs together every observed conformation of that sequence, so that all the crystallographic evidence for the loop is retained, and it additionally ORs in 100 computed conformers of the same capped molecule. The result is a description of what that peptide *can* present, anchored on what it has been seen to present. Because both sources contribute, the effective conformer count exceeds one hundred. The observed geometries are never discarded in favor of the computed ones; the computed ones fill in the accessible space around them.

Repeated sequences are kept rather than deduplicated. How often a loop sequence appears, and in how many distinct geometries, is information about how that sequence actually occupies space, and collapsing it to one representative would discard exactly the evidence the corpus was built to carry.

Peptide ensembles use the identical fingerprint convention and the same conformer recipe as the screening collection, which is the only reason the two chemistries can be trained and evaluated in one model.

2.4 Model

The input is a 2048-bit binary Morgan fingerprint of radius 2 [6], a connectivity descriptor computed from the two-dimensional structure alone. The network is a feedforward multilayer perceptron with two hidden layers of 1024 and 512 units and a 10,549-unit sigmoid output, one unit per pharmacophore bit, trained with binary cross-entropy. Predicted bits are thresholded at 0.5. No three-dimensional information of any kind enters the model at training or inference time.

2.5 Evaluation protocol

Evaluation molecules are those fingerprinted after the training snapshot was taken, so the test set is defined by time rather than by sampling: 155,648 screening collection molecules, fingerprinted after this model's corpus was sealed. That is the collection only population behind the per molecule fidelity figures in this section. Where all three corpora are scored together the post snapshot population is 51,291 molecules: 38,185 collection, 7,406 ChEMBL below 600 Da, 3,200 ChEMBL above 600 Da and 2,500 loop peptides. The two counts describe different populations, not alternative measurements of the same one. Three quantities are reported. Per-molecule fidelity is the Matthews correlation between the predicted and reference bit vectors. Pairwise fidelity is the agreement between the Tanimoto similarity computed from two predicted fingerprints and the same quantity computed from the two reference fingerprints, over 4,999,969 pairs. Ranking accuracy is the fraction of candidate pairs placed in the correct order, reported as a function of the true similarity difference, since pairs separated by less than the calculation's own noise are not meaningfully rankable by anything.

2.6 Domain of applicability

A critical assumption underlies every result reported here, and it defines what the model is entitled to be asked. Every molecule used for training and for evaluation is one that has actually been made or actually observed: compounds held in a commercial screening collection, and peptides extracted from experimentally determined protein structures. The model has therefore only ever been shown chemistry that is synthetically real, and a degree of synthetic feasibility is built into its training distribution rather than imposed on it afterwards.

An arbitrary two-dimensional structure that has never been synthesised is consequently outside the domain the model was fitted on. Such a molecule is not necessarily excluded, and the model will return an answer for it, but that answer is an extrapolation and should be treated as one. This also disposes of a control that would otherwise be conventional. Comparing against a predictor that has been shown no ligand at all presumes that molecules can be drawn from some unconstrained space and fed to the model, which is not the setting: the relevant question is not whether the model beats knowing nothing, but whether it is doing something a cheap two-dimensional descriptor could already do. Section 3.2 answers that question directly.

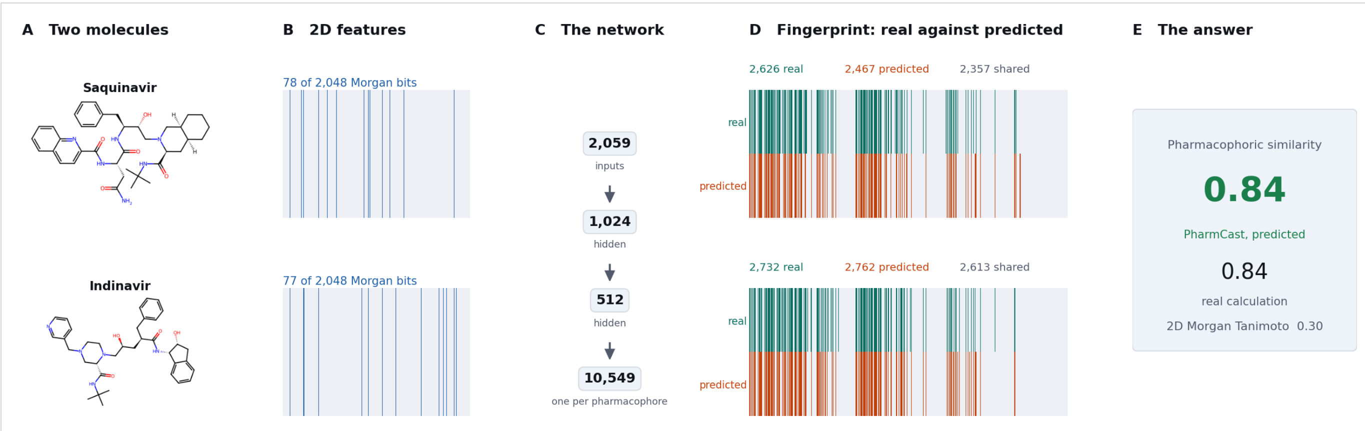

Figure 4. Two molecules carried through every stage. Panel A, the input structures. Panel B, the 2D feature vectors that are the model's only input. Panel C, the network. Panel D, each predicted fingerprint against its reference on the same bit axis. Panel E, the pharmacophoric similarity, which is the quantity actually consumed downstream.

3. Results

3.1 Fidelity

Accuracy is reported on three populations the model never trained on, and they are not interchangeable. The **reserved peptide test set** is the one to read first: 13,500 loops held out of the corpus by construction and never trained on, at a median per-molecule agreement of 0.914.

The catalog and ChEMBL populations are held out by TIME, every molecule fingerprinted after the corpus was sealed. On 155,648 such catalog molecules the median per-molecule agreement is 0.881. **Read that n before reading that number.** Catalog fingerprinting is now essentially complete, so what remains outside the training snapshot is the tail of a finished build rather than a sample of catalog chemistry. The pairwise statistics on the same molecules are the strongest of any population, at a median absolute error of 0.008 and Pearson 0.980; agreement per molecule falls while error improves, which is the signature of a changed population rather than a worse model.

The quantity that matters more is the similarity between two molecules. Across 4,999,969 held-out pairs the predicted and reference similarities agree at Pearson 0.980 and Spearman 0.977, with a median absolute error of 0.008 (Figure 2).

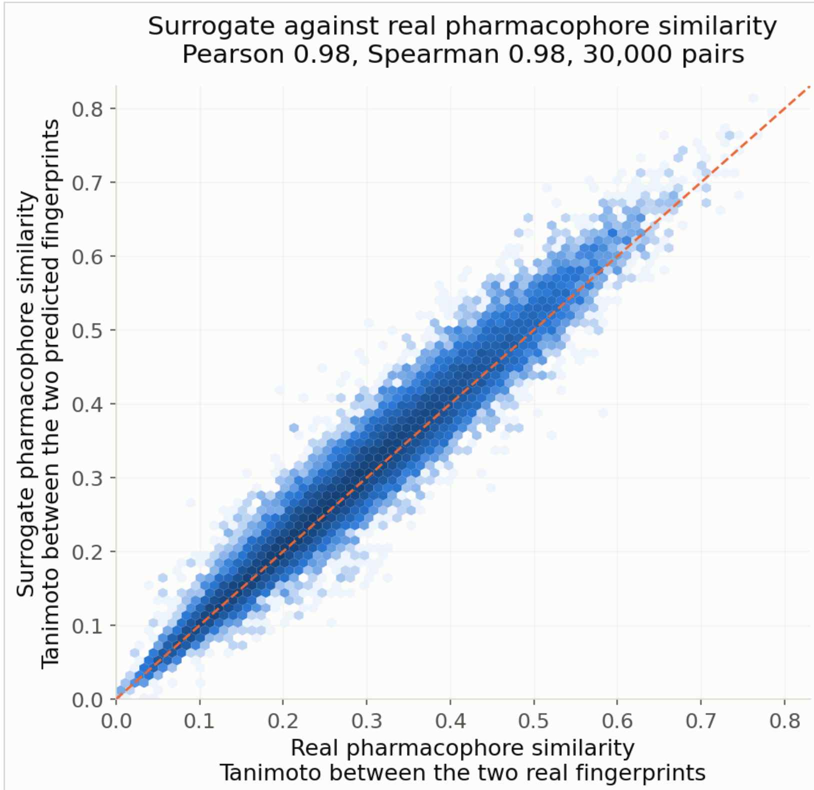

Figure 5. Predicted against reference pharmacophoric similarity for 4,999,969 held-out pairs. The diagonal marks perfect agreement.

3.2 The model is not re-deriving two-dimensional similarity

Since the input is a two-dimensional descriptor, the obvious concern is that the model has learned a monotone function of two-dimensional similarity and is adding nothing three-dimensional. It has not. Across the held-out pairs, the Morgan similarity of a pair correlates with the reference pharmacophoric similarity of the same pair at only 0.42, and the best straight-line predictor of pharmacophoric similarity from Morgan similarity alone carries a median absolute error of 0.065, against PharmCast's 0.008.

The decisive observation is that accuracy does not depend on how alike the two molecules look on paper (Table 2). Pairs that are essentially unrelated in two dimensions are predicted as accurately as pairs that share obvious features. That is the property scaffold hopping requires, and a model that had merely learned two-dimensional similarity could not have it. It is worth adding that the held-out pairs are overwhelmingly dissimilar to begin with: the median pair scores 0.09 and the 99th percentile 0.23, so within this collection "more alike" means "less unlike".

Table 2. Prediction accuracy as a function of the two-dimensional Morgan similarity of the pair being compared.

MORGAN SIMILARITY OF THE PAIR	PAIRS	MEDIAN ERROR	CORRELATION
0.00 to 0.10	92,660	0.006	0.964
0.10 to 0.20	67,321	0.012	0.983
0.20 to 0.30	3,610	0.022	0.975
0.30 to 0.50	271	0.028	0.965

3.3 Ranking and retrieval

Most applications consume an order rather than a value. Table 1 reports ranking accuracy as a function of the true similarity difference between the two candidates being compared.

Table 3. Ranking accuracy against the true similarity difference, over 2,998,976 comparisons drawn from the held-out set.

TRUE DIFFERENCE	COMPARISONS	ORDERED CORRECTLY
any	2,998,976	94.1%
greater than 0.05	2,308,250	99.0%
greater than 0.10	1,672,454	99.71%
greater than 0.20	719,525	99.96%

In a retrieval setting over 22,793 candidates and 22,793 queries, the true nearest neighbor has a median rank of 5 under the surrogate, and appears within the surrogate's top 100 for 89.3% of queries and within its top 500 for 97.5% (Figure 6). This is the operationally relevant behavior: the surrogate is used to select a shortlist that the reference calculation then adjudicates.

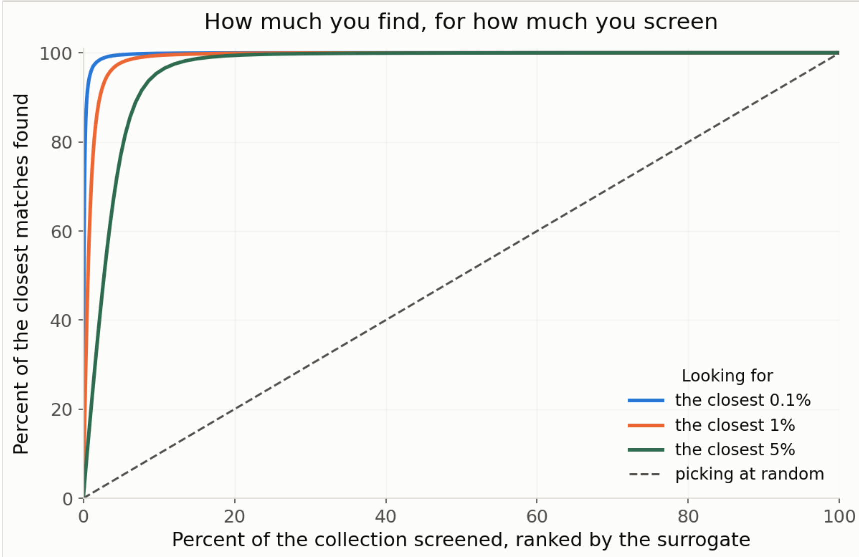

Figure 6. Fraction of the genuinely closest chemistry recovered after screening a given fraction of a collection by surrogate score. Left, full range; right, the same curves on a logarithmic depth axis, where selection decisions are made. The dashed line is screening in arbitrary order.

3.4 Throughput

Table 4 gives the cost of each route. The surrogate removes the conformational stage entirely rather than accelerating it, which is why the speedup is of a different order to what is normally achievable by optimization.

Table 4. What one pharmacophore fingerprint costs, and what the whole catalog costs, by route.

ROUTE	ONE FINGERPRINT	THE 4.65 MILLION COMPOUND COLLECTION
Conformer ensemble plus reference calculation	2.86 s	3,694 core hours
PharmCast from SMILES	**0.288 ms**	**29.3 minutes**

Conditions. All timings here were measured on one machine, an Apple M3 Ultra with 28 CPU cores, a 60 core GPU and 256 GB of unified memory, running macOS 26.5.1 with PyTorch 2.9.1 on CPU. The GPU was not used, and both routes were timed on the same hardware, so the comparison is like for like. Per molecule cost depends on the batch regime as well as on the machine: 0.288 ms in a library scale batch against 0.357 ms for a molecule processed alone. The 3,694 core hour figure is extrapolated from the per molecule rate, because the collection has been fingerprinted by that route only in multiday production runs; the 29.3 minute figure was measured end to end at 2,644 compounds per second on six worker processes, including catalog reading and featurisation. A timing quoted without these conditions is not reproducible, which is why they are stated here rather than in a footnote.

Of the conventional 2.86 s, 2.82 s is conformer generation and 0.039 s the fingerprint calculation over the ensemble. Comparing two molecules once both fingerprints exist is a Tanimoto over 10,549 bits and costs 0.012 ms by either route, so it is not where the difference lies. Starting from two SMILES strings, PharmCast returns a similarity in 0.584 ms at library scale against 5.71 s for the conventional route. The saving is not an optimization of the conventional route but the removal of its dominant stage: PharmCast never generates a conformer and never fingerprints one.

3.5 Behavior across chemistries

Catalog chemistry is not the only domain of interest. Table 5 reports the composite model pharmcast_scp_v10.pt, trained on **4,563,262** collection molecules together with **121,784** ensemble-enhanced protein loop peptides, evaluated on held-out pairs from three chemistries: the screening collection, loop peptides whose sequences the model never saw, and real ChEMBL compounds above 600 molecular weight, each carrying a measured activity value, which lie outside the size range the collection covers (Figure 7).

Table 5. Agreement with the reference calculation across three chemistries, 1,200 held-out pairs each.

TEST CHEMISTRY	MEDIAN ERROR	CORRELATION
Screening collection	0.008	0.978
Protein loop peptides	0.016	0.985
Large ChEMBL compounds	0.027	0.937
All three combined	**0.016**	**0.980**

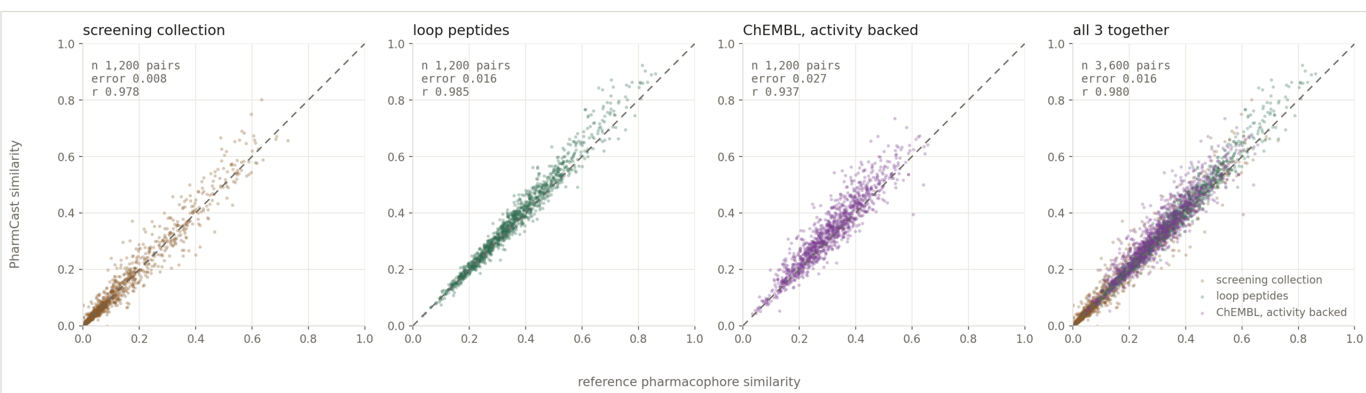

Figure 7. Reference against predicted similarity on held-out pairs from three chemistries and then all together. The combined panel should be read as a statement about coverage rather than as a single accuracy, since it blends two chemistries the model handles well with one it handles coarsely.

3.6 The limit: molecular size

The clearest failure mode is size. Table 6 compares performance on catalog chemistry against real ChEMBL compounds above 600 molecular weight, using an identical protocol on both. The two similarity landscapes have essentially the same spread, 0.153 against 0.118, so the degradation is attributable to size rather than to one task being intrinsically harder.

Table 6. The same protocol applied to catalog chemistry, median 211 molecular weight, and to 573 real compounds above 600, median 638. The two similarity landscapes have comparable spread, 0.153 and 0.118, so the difference is attributable to size rather than to one task being intrinsically harder.

MODEL	CATALOG ACCURACY	CATALOG ERROR	LARGE ACCURACY	LARGE ERROR
PharmCast-SCP v10	**93.4%**	**0.008**	**82.6%**	**0.040**

Ranking accuracy on the larger compounds sits below the catalog figure: 82.6% against 93.4%, at a median absolute error of 0.040 against 0.008. That gap is the honest limit of the model on chemistry heavier than the screening collection contains, and it narrows as the activity-backed ChEMBL corpus fills the 142 to 1000 Da range.

3.7 What the reference calculation itself can support

Any surrogate is bounded by the reproducibility of the quantity it approximates. We measured this directly by rebuilding 149 of the large molecules with a different conformer embedding seed, changing nothing else, and comparing the two independent ground truths. Pairwise similarity reproduces to a median absolute difference of 0.006 at Pearson 0.995. The individual fingerprints are noticeably less reproducible than the similarities computed from them, at a median self-similarity of 0.937 and a median self-agreement of 0.958, which indicates that the conformer-to-conformer variation is largely common-mode and cancels in a comparison.

This bounds the interpretation of Section 3.6. The reference calculation agrees with itself an order of magnitude more tightly than PharmCast agrees with it on large molecules, so that degradation is a genuine limitation of the surrogate and not an artifact of a noisy target.

4. Discussion

The intended use follows from the measurements. Where the true difference between two candidates is large, the surrogate orders them almost perfectly and can be trusted to build a shortlist. Where the difference is small, it cannot, and the reference calculation must adjudicate. The economics are favorable because the two regimes are cheap and expensive respectively: a surrogate pass over a multi-million compound collection costs minutes, and the reference calculation is then spent only where it decides the answer.

One failure mode deserves emphasis because it is not visible in any of the tables above. When a surrogate is used as the objective of an optimizer rather than as a filter, the optimizer will eventually exploit its error. In a design campaign that used this model as the scoring function for a genetic algorithm, a seventeen-fold increase in search budget raised the median surrogate score of the retained designs by 0.20 while raising their median true score by 0.02, and the surrogate's own median error over those designs rose from +0.06 to +0.24. The search had become better at finding molecules the model was wrong about. The practical consequences are that the surrogate should be used to rank and never to decide, that a real rescore is a required stage rather than a formality, and that many survivors rather than a top handful should be carried into it.

5. Limitations

The model predicts the ORed ensemble fingerprint and therefore cannot support any application that requires per-conformer matching geometry, which is the regime PolyPharmPrint addresses. Performance degrades materially outside catalog-like chemistry, most clearly with increasing molecular size, and the degradation is a property of the surrogate rather than of a noisy target. Predicted similarities carry a small positive bias arising from a systematic over-prediction of set bits; this does not affect ordering but does affect any absolute threshold. All results reported here are computed on real catalog compounds and on peptides extracted from experimentally determined structures; no molecules were generated for evaluation purposes.

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Data and provenance

Every number in this document is measured on molecules the model was not trained on, and each is drawn from a named evaluation file rather than transcribed. The three evaluation populations are distinct and are never pooled into a single headline:

Table 8. What each reported population is, and how it was held out. Counts are for the model reported here.

POPULATION	HOW IT IS HELD OUT	SOURCE
Catalog chemistry	Every collection molecule fingerprinted *after* the training corpus was sealed, so none could have been seen. Held out by time, not by sampling.	[8]
ChEMBL, activity backed	Every ChEMBL molecule fingerprinted after the seal, each carrying a measured activity value.	[11]
Loop peptides	A dedicated test set, never trained on: structures at resolution 1.8 Å or better and free R-factor 0.23 or better, excluding every entry used to build the training corpus *and* every loop whose amino acid **sequence** already appears in it. Novel by sequence rather than merely by canonical structure, and better resolved than the training corpus, so a poor score here could not be attributed to soft coordinates.	[10]

The peptide distinction matters because the loop corpus is nearly complete. What remains outside it is the residue of a finished build rather than a sample of peptide chemistry, and scoring on that residue understates the model substantially. The dedicated test set is what the peptide figures here are measured on.

All figures and tables in this document are generated from the evaluation files at build time; no value is transcribed by hand. The conformer protocol, the fixed seed and the fingerprint width are identical for every training and every evaluation molecule.